MILITARY SPECIFICATION

CARTRIDGE, 105MM, TNT, HE, M760
LOADING, ASSEMBLING, AND PACKING

This specification is approved for use by the U.S. Army Armament Research and Development Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the loading, assembling, and packing for one type of HE, TNT, Single Zoned Cartridge designated as M760 (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-P-46565	-Primer Percusion, M28A2 and M28B2, Loading, Assembling and Packing.
MIL-A-48078	-Ammunition, Standard Quality Assurance Provisions, General Specification for

STANDARDS

MILITARY

MIL-STD-105	-Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-453	-Inspection, Radiographic
MIL-STD-1235	-Single and Multilevel Continuous Sampling Procedures and Tables for by Attributes.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, Attn. DRDAR-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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DRAWINGS (see 6.3)

U.S. ARMY ARMAMENT RESEARCH AND DEVELOPMENT COMMAND

9211610-2	-Projectile, 105MM, HE, M1 Loading Assembly (TNT Charge)
9278812-2	-Projectile, 105MM, HE, M1, Loading Assembly (TNT Charge)
9289185	-Cartridge, 105MM, HE, M760, with M200 Propelling Charge Assembly for Howitzer, M204, Loading

PRODUCT AND PACKAGE DRAWINGS

7549072	-Box, Packing, Ammunition in Fiber Container M105A2
7549073	-Container, Ammo, Fiber, M105A2 for 105MM Howitzer Ammunition.
9258026	-Container, Ammo, Fiber, PA55, for 105MM Howitzer Ammunition without Fuze.
9258027	-Box, Packing, Ammo for 105MM Howitzer, Ammo, w/o Fuze in Fiber Container PA55.
9271850	-Packing and Marking for Box, Wirebound, Packing Ammunition without Fuze in Fiber Container PA55.
9275180	-Marking for Box, Wirebound, Packing, Ammunition for 105MM Howitzer Ammunition.
9275181	-Packing and Marking for Box, Wirebound Packing, Ammunition for 105MM, Howitzer Ammunition in Fiber Container, M105A3.

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

3. REQUIREMENTS

3.1 Material. Material and parts shall be in accordance with applicable drawings and specifications (see 6.4).

3.1.1 Clean explosive scrap. Clean explosive scrap produced in regular operations may be remelted and reused provided that it complies with the requirements for the grade of

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high explosive specified, except for form. Scrap from floor sweeping, ventilation and suction apparatus (other than from drilling operations specifically controlled to prevent contamination of explosive) shall not be used (see 6.5).

3.2 Cartridges. The cartridges shall comply with all requirements specified on Drawing (dwg.) 9289185 and with all requirements specified in applicable specifications.

3.3 Cavitation in high explosive charge.

3.3.1 Projected cavities. Cavities having a projected length of 1/32 inch (in.) or less shall be disregarded. Cavities within the explosive charge shall not exceed the requirements specified in Table I, when determined as specified in 4.4.1.

TABLE I

(Segments see Figure I)

	A	B	C	D
Sum of projected areas of the cavities, excluding pipes, cracks and annular rings, square inch (sq. in.).	*1/64	1/4	1/2	1/2
Projected length of any cavity, excluding pipes, cracks, and annular rings (in.).	1/8	1/2	1/2	3/4
Piping cavities, maximum (max.) projected area (sq. in.).	0	1/4	1/2	--
Piping cavities, max. projected width (in.).	0	1/4	1/4	--
Cracks, max. projected width (in.).	1/32	1/32	1/32	--
Annular rings, max. projected width (in.).	0	0	1/4	--

*If the length of the largest cavity is 1/16 inch or less, the maximum total projected area may be 1/20 square inch. Piping cavitation is defined as cavitation located on or near the vertical center line of the projectile with the longitudinal axis (length) three times or greater than the width.

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3.3.2 Surface cavitation. Cavities formed due to chipped or broken explosive from the sidewalls of the fuze well shall not extend in aggregate around more than 1/4 of the circumference. Cavities in the base of the fuze well shall not be cause for rejection provided the sum of the areas of the cavities is not greater than 80 percent of the area of the fuze well base and no individual point extends more than 1/4 inch below the maximum depth of the fuze well permitted by the drawing, and provided no point extends above the flat surface on which the liner would rest. (Cavities having a maximum dimension of 1/32 inch or less shall be disregarded in this computation as shall voids within the applicable forward region of the fuze well prescribed by dwg. 9211610-2 or dwg. 9278812-2).

3.3.3 Porous areas. Porous areas shall be treated as cavities and shall be subject to the restrictions placed upon cavities, except that 80 percent of the projected length and 80 percent of the projected area shall be considered for acceptance purposes.

3.3.4 Cracked charges. Not more than two transverse cracks shall be permitted in any charge, and not more than one transverse crack shall be permitted in Segment A.

3.3.5 Base separation. There shall be no separation between the shell base and explosive charge.

3.4 Proving ground test. The cartridge shall comply with the following requirements when tested as specified in 4.4.2.

3.4.1 Premature burst and security of parts. There shall be no premature burst or indication of metal parts failure in the gun bore or in flight before impact.

3.4.2 Safety. There shall be no premature burst or indication of metal parts separation of the cartridge in the gun bore or in flight when tested as specified in 4.4.2.1, 4.4.2.2 and 4.4.2.3.

3.4.3 Functioning/precision

3.4.3.1 Functioning. The fuze shall cause the projectile to function high order.

3.4.3.2 Precision

3.4.3.2.1 Pooled range probable error (applicable to M572 Fuze only). The pooled range probable error (RPE) of the ballistic samples fired shall not exceed the appropriate value listed below:

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Degrees of freedom	Maximum RPE (percent)
29	.327
28	.328
27	.330
26	.331
25	.333
19	.346
18	.348
17	.351
16	.354
15	.357

3.4.3.2.2 Horizontal Range. No projectile shall have a horizontal range less than 95 percent of the mean horizontal range.

3.4.3.3 Muzzle velocity. The muzzle velocity for the cartridge, fired at 70°F, shall not be more than 2195 feet per second nor less than 2145 feet per second and the standard deviation of the lot shall not be greater than 8 feet per second (1.22 meters per second).

3.4.3.4 Pressure. The pressure of the cartridge, fired at 70°F, shall not produce a chamber pressure greater than 47,000 psi.

3.5 Radiographic film. Only a film type and vendor as listed herein has been tested and approved by the U.S. Army Armament Research and Development Command for use in radiographing this projectile. A substitute item shall not be used before prior testing and/or approval by the U.S. Army Armament Research and Development Command, Dover, New Jersey 07801.

Eastman Kodak Company
1187 Ridge Road, West
Rochester, New York 14650

Film Types: AA, T, or M

E. I. DuPont de Nemours and Company
Photo Products Department
Wilmington, Delaware 19898

Film Types: NDT-55, NDT-65, or NDT-70

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3.6 Workmanship.

3.6.1 Cartridge. All manufacturing processing and assembly operations of the cartridge shall be correctly performed in a thoroughly workmanlike manner. Parts shall be free of burrs, chips, dirt cracks, grease, rust and other foreign material. The cleaning method shall not be injurious to any of the parts nor shall the parts be contaminated by cleaning agents.

3.6.1.1 Loading of high explosive. The interior surface of the projectile shall be dry when the high explosive is poured into the projectile. Solid explosives shall not be present in the projectile before loading or be added to the projectile during pouring or cooling.

3.6.1.2 Painting and marking. The painting and marking shall be restored by the assembling contractor on any components furnished to him painted and marked when such painting and marking has been marred or obliterated.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsiblity for inspection and standard quality assurance provisions. Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are here made a part of this detail specification.

4.2 Classification of inspections. The following type of inspection shall be conducted on this item:

a. Quality Conformance Inspection

4.3 Quality conformance inspection.

4.3.1 Inspection lot formation. Inspection lots shall comply with the lot formation provisions of MIL-A-48078. In addition, inspection lots of cartridges shall contain:

a. Cartridge cases of one lot interfix number from one manufacturer.
b. Projectile metal parts of one lot interfix number from one manufacturer.
c. Propelling charges from not more than one lot.
d. Loaded primers of one lot interfix number from one manufacturer.
e. Supplementary charges from one manufacturer produced in accordance with the same drawing, same drawing revision, same specification and same specification revision.

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f. Fuze well liners from one manufacturer produced in accordance with the same drawing, same drawing revision, same specification and same specification revision.

g. TNT of the same type from one lot interfix number from one manufacturer.

h. Loaded projectile of same weight zone.

i. Loaded fuzes from one lot interfix number from one manufacturer.

4.3.2 Examination. See MIL-A-48078.

a. Sampling plans. Unless otherwise specified in the Classification of Defects and Test tables, sampling plans and procedures for major and minor defects shall be in accordance with MIL-STD-105, Inspection Level II.

The contractor may elect to perform 100% inspection with automated equipment in lieu of sampling inspection for defects listed in this specification. In such cases, calibration plans shall be submitted to the Technical Agency (see 6.15) for approval for each automatic inspection station. The plans shall identify: (1) the frequency (production interval) for validating equipment functioning (2) the methods and procedures for validation including designs of defect standards and (3) a description of material control procedures to be imposed on a product passing through each automated station during the production interval between validations. Under these provisions final acceptance will be withheld from material produced during a production interval until satisfactory validation of each of the automated stations so controlled. In the event of a validation failure of any station, the material processed through such station since the last acceptance validation shall be placed in a rejected status pending corrective action. Under certain circumstances, and for sufficient cause, the contractor may request the government's permission to use process controls or process inspections in lieu of the product inspections contained herein. In such instances, process control plans containing the aforementioned elements shall be submitted to the technical agency for approval.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-63261 (AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER 9211610 -2 or 9278812 -2 as appl.
4.3.2.1	Projectile, loading assembly, prior to loading explosive		NEXT HIGHER ASSEMBLY 9289185

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1.	Foreign matter in charge cavity		100%	3.2	Visual
2.	Unsolidified pool of paint in bottom of cavity.		100%	3.2	Visual
Major					
101.	Interior protection coating with bare spot or blister.		0.40%	3.2	Visual
Minor					
201.	Evidence of poor workmanship		0.65%	3.6	Visual

NOTES:

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MIL-C-63261 (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.3.2.2	Projectile loading assembly, prior to assembly of liner.	1 OF 1	9211610 -2 or 9278812 -2 as appl.
			NEXT HIGHER ASSEMBLY 9289185

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1.	Explosive in threads		100%	3.2	Visual
Major					
101	Depth of fuze well, maximum (max.)		0.40%	3.2	Gage/CD
102	Foreign matter in fuze well		0.40%	3.2	Visual
Minor					
201	Diameter of fuze well, max.		0.65%	3.2	Gage/CD
202	Surface cavitation, excessive		0.65%	3.3.2	Visual
203	Evidence of poor workmanship		0.65%	3.6	Visual

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PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.3.2.3	Projectile loading assembly, prior to assembly of supplementary charge		9211610 -2
			NEXT HIGHER ASSEMBLY 9289185

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1.	Liner missing		100%	3.2	Visual
Special	(see 6.12)				
a.	Explosive on exterior of projectile		100%	3.2	Visual
Major					
101.	Concentricity of inside diameter of liner with pitch diameter of thread including perpendicularity with face of body.		0.40%	3.2	Gage/CD
102.	Depth to bottom of liner, minimum		0.40%	3.2	Gage/CD
Minor					
201.	Liner removable by hand		0.65%	3.2	Manual
202.	Liner damaged.		0.65%	3.2	Visual
203.	Foreign matter, including explosive, in-liner cavity		0.65%	3.2	Visual
204.	Evidence of poor workmanship		0.65%	3.6	Visual

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PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.3.2.4	Projectile loading assembly, prior to assembly of closing plug.		9211610 -2
			NEXT HIGHER ASSEMBLY 9289185

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1.	Pad on supplementary charge missing		100%	3.2	Visual
2.	Supplementary charge missing		100%	3.2	Visual
3.	Projectile zone weight incorrect by two or more zones.		100%	3.2	Balance
4.	Explosive on exterior of projectile		100%	3.2	Visual
Major					
101.	Weight of loaded shell incorrect by one zone with supplementary charge & liner. but without a spacer and closing plug.		0.40%	3.2	Balance
102.	Supplementary charge not removable by hand.		0.40%	3.2	Manual
103.	Weight zone marking incorrect or unidentifable		0.40%	3.2	Visual
104.	Supplementary charge inverted		0.40%	3.2	Visual
Minor					
201.	Supplementary charge damaged		0.65%	3.2	Visual
202.	Spacer.missing or damaged		0.65%	3.2	Visual
203.	Rotating band damaged		0.65%	3.2	Visual
204.	Protective coating with total damage area in excess of $\frac{1}{4}$ square inch		0.65%	3.2	Visual
205.	Marking missing, misleading or unidentifiable (except weight zone marking)		0.65%	3.2	Visual
206.	Evidence of poor workmanship		0.65%	3.6	Visual

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PARAGRAPH	TITLE	SHEET 1 OF 2	DRAWING NUMBER
4.3.2.5	Projectile loading assembly, prior to assembly of closing plug		9278812-2
			NEXT HIGHER ASSEMBLY 9289185

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1.	Liner missing		100%	3.2	Visual
2.	Projectile zone weight incorrect by two or more zones.		100%	3.2	Balance
3.	Explosive on exterior of projectile		100%	3.2	Visual
Major					
101.	True position of inside diameter of liner with face of body and pitch diameter of threads.		0.40%	3.2	Gage/CD
102.	Depth to bottom of liner, min.		0.40%	3.2	Gage/CD
103.	Weight of loaded shell incorrect by one zone with liner, but without a spacer and a closing plug		0.40%	3.2	Balance
104.	Weight zone marking incorrect		0.40%	3.2	Visual
Minor					
201.	Liner removable by hand		0.65%	3.2	Manual
202.	Liner damaged		0.65%	3.2	Visual
203.	Foreign matter, including explosive, in liner cavity		0.65%	3.2	Visual

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PARAGRAPH	TITLE		SHEET 2 OF 2		DRAWING NUMBER
4.3.2.5	Projectile loading assembly, prior to assembly of closing plug				9278812-2
					NEXT HIGHER ASSEMBLY 9289185
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
204.	Rotating band damaged		0.65%	3.2	Visual
205.	Marking (other than weight zone marking), missing, misleading or unidentifiable.		0.65%	3.2	Visual
206.	Evidence of poor workmanship		0.65%	3.6	Visual

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PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.3.2.6	Primer, prior to assembly to case				8838130
					NEXT HIGHER ASSEMBLY 9289185
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical 1.	One or more flash holes missing		100%	3.2	Visual
Major	None defined				
Minor	None defined				

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PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.3.2.7	Cartridge case, prior to assembling propelling charge		9289185
			NEXT HIGHER ASSEMBLY: NA

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101.	Primer missing or damaged		0.40%	3.2	Visual
Minor					
201.	Foreign matter in interior of case		0.65%	3.2	Visual
202.	Evidence of poor workmanship		0.65%	3.6	Visual

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PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.3.2. 8	Propelling charge, prior to assembly to cartridge case.		9282042
			NEXT HIGHER ASSEMBLY 9289185

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major 101.	Any opening in bag large enough to permit escape of propellent		0.40%	3.2	Visual
Minor 201.	Evidence of poor workmanship		0.65%	3.6	Visual

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PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9289185
4.3.2.9	Cartridge case prior to assembling to projectile				NEXT HIGHER ASSEMBLY NA
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major 101.	Propelling charge missing		100%	3.2	Visual
Minor	None defined				

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PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.3.2.10	Cartridge assembly				9289189
					NEXT HIGHER ASSEMBLY
					NA
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1.	Primer above flush		100%	3.2	Gage/Gb
Major					
101.	Component missing		0.40%	3.2	Visual
Minor					
201.	Exterior coating of shell with an aggregate of bare spot in excess of $\frac{1}{4}$ sq. in.		0.65%	3.2	Visual
202.	Plug not fully seated		0.65%	3.2	Manual
203.	Marking missing, unidentifiable, misleading or incorrect color (except zone weight marking.)		0.65%	3.2	Visual
204.	Evidence of poor workmanship		0.65%	3.6	Visual

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PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.3.2.1.1	Unsealed fiber container		7549073 or 9258026
			NEXT HIGHER ASSEMBLY: NA

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Packing component missing		0.40%	5.1	Visual
Minor					
201	Cartridge cannot be removed by hand		0.65%	5.1	Manual
202	Container cap cannot be removed by hand		0.65%	5.1	Manual
203	Glue or asphalt on cartridge		0.65%	5.1	Visual
204	Packing components improperly assembled		0.65%	5.1	Visual

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PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.3.2.12	Sealed fiber container				7549073 or 9258026
					NEXT HIGHER ASSEMBLY
					NA
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Container cut or damaged through all asphalt impregnated layers		0.40%	5.1	Visual-Manual
102	Metal end loose or distorted		0.40%	5.1	Visual-Manual
103	Tape incomplete or badly wrinkled		0.40%	5.1	Visual
Minor					
201	Tear tab length, min.		0.65%	5.1	Gage/CD
202	Contents loose		0.65%	5.1	Manual
203	Gap between cover and body container exceeding 1/8 inch		0.65%	5.1	Visual
204	Cuts, scuffs or gouges in outer layers.		0.65%	5.1	Visual
205	Marking misleading, missing or unidentiafiable.		0.65%	5.3	Visual

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MIL-C-63261 (AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.3.2.13	Sealed wood packing box		7549072, 9258027, 9271850 or 9275181 as applicab
			NEXT HIGHER ASSEMBLY NA

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Box damaged exposing contents		0.40%	5.2	Visual
102	Hardware or strapping missing, broken or loose		0.40%	5.2	Visual-Manual
103	DOT symbol missing or incorrect		0.40%	5.2	Visual
104	Board broken		0.40%	5.2	Visual
Minor					
201	Strapping or hardware improperly assembled		0.65%	5.2	Visual-Manual
202	Handle missing or insecure		0.65%	5.2	Visual-Manual
203	Car seal missing, unsealed or improperly positioned		0.65%	5.2	Visual
204	Marking (other than DOT) misleading, missing or unidentifiable		0.65%	5.3	Visual
205	Contents loose		0.65%	5.2	Manual

NOTES:

DLA AR-QA Form 160 Jul 77 Replaces SARPA-QA Form 2567 Feb 74 Which is Obsolete

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4.3.3 Testing

4.3.3.1 Check test for possible deterioration of primer percussion, M28 (see specification MIL-P-46565). If the total elasped time between original acceptance of the M28 primer lot and the assembly of that lot into complete round cartridges exceeds two years, or if the M28 primers have been subjected to adverse conditions, however brief, at any time since previous tests, the applicable lot shall be subjected to and must satisfactorily pass the check test for deterioration specified in Specification MIL-P-46565 immediately before the M28 primer lot is assembled into the cartridges. The test shall be performed by the contractor on assemblies selected by the Government inspector at the facility assembling the complete cartridges (see 6.6).

4.3.3.2 Cavitation (see 3.3, except 3.3.2). Cavitation defects are classified as follows:

Critical:

1. Cavitation, (including porous areas), cracks, or annular rings in Segment A, B, C, or D in excess of that permitted in Table I.
2. Cracks in excess of that permitted in 3.3.4.
3. Base separation in excess of that permitted by 3.3.5.

The sampling plan and procedures for this test shall be in accordance with MIL-STD-1235, using a CSP-1 plan having an AQL of 0.010 percent. The appropriate production interval shall be as specified by the procuring activity. The test shall be performed as specified in 4.4.1, using equipment in accordance with 4.3.4. The equipment needed for this test is CD.

4.3.3.3 Bourrelet diameter (see dwg. 9211610, or 9278812 as applicable, Major defect. CD. The projectile shall be gaged 100 percent for maximum diameter of bourrelet. Any projectile which fails to comply with the applicable requirement shall be classed defective and removed from the lot. Equipment shall be in accordance with 4.3.4. (Note: The inspection shall not be required if it is determined that the projectiles have been previously gaged 100 percent for this characteristic.) All projectiles retouched shall be gaged.

4.3.3.4 Fit of projectile with cartridge case (see dwg. 9289185) - Major defect. CD. The cartridge cases shall be gaged 100 percent for profile and diameter of the mouth to assure a proper fit with the projectile. Any case which fails to comply with the applicable requirements shall be classed defective and removed from the lot. Inspection equipment shall be in accordance with 4.3.4 (Note: This inspection shall not be required if it is determined that the cartridge cases have been previously gaged 100 percent for these characteristics).

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4.3.3.5 Proving ground test.

4.3.3.5.1 Premature bursts and security of parts (see 3.4.1), Critical defect. If any of the sample projectiles used in any of the tests fails to comply with the applicable requirements the lot shall be rejected.

4.3.3.5.2 Safety (see 3.4.2), Critical defect.

4.3.3.5.2.1 Beginning with the first lot produced and continuing until three consecutive lots have complied with the applicable requirements, twenty sample projectiles shall be randomly selected from each lot for this test. The projectiles shall be assembled with accepted M572 Fuzes. If any projectile fails to comply with the requirements of 3.4.2, when tested as specified in 4.4.2.2, the lot shall be rejected.

4.3.3.5.2.2 After three consecutive lots have met the criteria of 4.3.3.5.2.1, twelve sample projectiles shall be randomly selected from each lot for this test. The projectiles shall be assembled with accepted M572 Fuzes. If any projectile fails to comply with the requirements, the lot shall be rejected and lot sampling again shall be in accordance with 4.3.3.5.2.1.

4.3.3.5.3 Functioning/precision, Fuze, M572 (see 3.4.3), Major defect. Beginning with the first lot produced and continuing until three consecutive lots have complied with the applicable requirements, thirty (30) sample projectiles shall be randomly selected from each lot for this test. The projectiles shall be assembled with accepted M572 Fuzes. If any projectile fails to comply with 3.4.1 or if four or more projectiles fail to comply with 3.4.3.1 or if the projectiles fail to comply with 3.4.3.2, when tested as specified in 4.4.2.1, the lot shall be rejected. After three consecutive lots have complied with the applicable requirements, this test shall be discontinued.

4.3.3.5.4 Muzzle velocity /Functioning with fuze, M572 (see 3.4.3.3) - Major defect. Beginning with the first lot produced and continuing until three consecutive lots have complied with the applicable requirements specified, the sampling plan shall be in accordance with MIL-STD-414, Section B, Table B-3, Code Letter H and an AQL of 2.5 percent. In addition, if the standard deviation of the sample multiplied by the factor (see 6.17) exceeds the applicable lot requirement, when tested as specified in 4.4.2.3, the lot shall be rejected. Also, if any projectile fails to comply with 3.4.1 or if two or more projectiles fail to comply with 3.4.3.1, the lot shall be rejected.

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4.3.3.5.4.1 After three consecutive lots have met the criteria of 4.3.3.5.4, the sampling plan shall be in accordance the MIL-STD-414, Section B, Table B-3, Code Letter F and an AQL of 2.5 percent. In addition, if the standard deviation of the sample multiplied by the factor (see 6.17) exceeds the applicable lot requirement, the lot shall be rejected. Also, if any projectile fails to comply with 3.4.1 or if two or more projectiles fail to comply with 3.4.3.1, the lot shall be rejected. If lot failure occurs with three lots in succession, the sampling provisions shall revert back to 4.3.3.5.4.

4.3.3.5.5 Pressure Test, with Fuze, M572 (see 3.4.3.4). Major Defect. The cartridges used in 4.3.3.5.4 shall be used for this test. Beginning with the first lot produced and continuing until three consecutive lots have complied with the applicable requirements specified, the sampling plan shall be in accordance with MIL-STD-414, Section B, Table B-1, Code Letter H and an AQL of 0.25%.

4.3.3.5.5.1 After three consecutive lots have met the acceptance criteria of 4.3.3.5.5, the cartridges used in 4.3.3.5.4.1 shall be used for this test. The sampling plan shall be in accordance with MIL-STD-414, Section B, Table B-1, Code Letter F and an AQL of 0.25%. If lot failure occurs with three lots in succession, the sampling provisions shall revert back to 4.3.3.5.5

4.3.4 Inspection equipment. The inspection equipment required to perform the examinations and tests prescribed, and noted as CD, is to be designed by the contractor. (See 6.7).

4.4 Test methods and procedures.

4.4.1 Determination of cavities. The sample projectiles shall be submitted to radiographic examination. The radiographic beam shall be located in a plane which is perpendicular to the longitudinal axis of the projectile and tangent to the base of its internal cavity. The outline of the explosive charge shall be clearly defined at all points. Examination of the radiograph shall be made to determine compliance with applicable requirements. No images of lead masks, containers, fixtures, identifying marks, or other extraneous features should project into the region of explosive under inspection, with the exception of specified penetrameters and radiation screening required to reduce film densities to readable levels in the thinner areas of the shell. Such screening should be free of any radiographically observable features in the projected area under inspection. Radiographic equipment, operators and procedures shall be qualified in accordance with MIL-STD-453 (see 6.8). In addition, the

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sensitivity of the radiographic technique shall be 2.0 percent and minimum photographic density of the film used shall be no less than 1.25 at every point within the projected area of explosive, including the base and rotating band area, with the maximum density no greater than to permit adequate inspection over the entire area of interest, as determined in accordance with 4.4.1.1 and 4.4.1.2. In the event an original radiograph leaves doubt as to its acceptability, the projectile shall be reradiographed 90 degrees from the original exposure. Acceptability shall be determined on the basis of the film giving the largest projected area of defect.

4.4.1.1 Check of sensitivity of radiographic technique. The sensitivity of radiographic technique shall be determined by means of penetrameters fabricated as specified herein. The penetrameters shall be placed on the radiographic source side of the projectile so that the 2T holes are at the points shown in Figure 2. Three penetrameters for each projectile shall be used. When radiographing a number of similar projectile the frequency of sensitivity check shall be no less than one in each fifty radiographs and on a rotating ring, the penetrameters shall be used at one station for each complete rotation of the ring. The sensitivity of the radiographic technique shall be considered satisfactory when the image of the penetrameters, including the 2T holes, are sharply defined on the radiograph.

The penetrameters shall be made of steel having an approximate specific gravity of 7.8 The penetrameters shall be located as shown on Figure 2. For further details on construction of the penetrameter see MIL-STD-453.

4.4.1.2 Check of photographic density of film. The photographic density of the film shall be determined on the film in the region of minimum film density within the projected area of the explosive charge. Film density shall be checked each hour of operation or fraction thereof, for each film processor. Greater frequency shall be required for any period during which difficulty is encountered in establishing the required densities. To judge densities it is recommended that standards be used (see 6.9). Photographic densities shall be measured with a diffuse density, photoelectric densitometer providing direct display of numerical density units. Accuracy of densitomer shall be maintained by calibrating with a photographic step tablet each hour immediately prior to measurements on the radiographic film. Calibration shall be made at a step tablet density level within .5D of that being measured on the radiographic film. Each point of density measurement on the

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radiographs shall be identified by a circumscribed circle (made with a suitable pen or pencil) of no more than 1/4 inch in diameter. All photographic step tablets used in normal operation shall be checked for accuracy monthly by calibrating the densitometer with a reference photographic step tablet calibrated by National Bureau of Standards or approved by ARRADCOM. Reference tablet shall be kept available, but shall be protected from wear and damage and shall not be used in daily operations. Any photographic step tablet having severe surface damage or showing a deviation from previous calibrations by .05D or more shall be replaced with a new, unused step tablet.

4.4.2 Proving ground. These tests shall be conducted at a Government proving ground in accordance with the applicable Acceptance Test Procedure. The sample cartridges shall be fired with M200 propelling charges in the M204 howitzer. The wear of the tube shall be no greater than 0.015 in. (0.38mm) when measured 25.25 in (641.3mm) from the rear face of the tube. Observation shall be made for conformance to all requirements.

4.4.2.1 Functioning/precision firings (w/M572 Fuze). All test and informational control projectiles shall be conditioned and fired at 70°F ± 2 1/2°F at Zone 8, using a M200 series propelling charge conditioned at 70°F ± 2 1/2°F. In order to minimize the effects of met conditions the test shall be fired in three series of 10 rounds each. The first two subgroups shall be fired at a quadrant elevation of 1146 Mils and the last subgroup shall be fired at a quadrant elevation of 381 Mils. Each 10 round subgroup shall be immediately preceeded by two (2) informational control rounds (see 6.14) and, in addition the final subgroup shall be followed by two (2) informational control samples. Time lapses of more than 15 minutes between consecutive rounds shall not be permitted and it is desirable that each 12 round series (2 informational control samples plus 10 test samples) be completed within one (1) hour. Range probable errors shall be calculated for each 10 round subgroup and then pooled range probable error shall be computed for determination of compliance with 3.4.3.2.1.

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$$\text{Subgroup RPE (percent) } RPEj = \sqrt{\frac{\Sigma X_i^2 - 1/n\ (\Sigma X_i)^2}{n-1}} \times \frac{100\% \times .6745}{\text{Ave. Range}(j)}$$

$$\text{Pooled RPE (percent)} = RPEp = \sqrt{\frac{RPE_1^2(n_1 - 1) + RPE_2^2\ (n_2 - 1) + \ldots + RPE_j^2(n_j - 1)}{DF}}$$

Where X = individual horizontal ranges within each subgroup.

Where DF = Degrees of freedom = $n_1 + n_2 \ldots . n(j)$ - number of subgroups.

Prior to computing the range probable errors each subgroup shall be evaluated for velocity outliers. To determine if an outlier exists the velocity values shall be placed in ascending or decending numerical order with the suspected value first.

V1, V2, V3-- Vi-1, Vi

Where V1 = Suspected outlier and Vi = Max. velocity if suspected outlier is low or min. velocity if suspected outlier is high.

From these values the statistic "r" shall be calculated and compared to values given. If "r" exceeds the table value the velocity shall be regarded as an outlier and the range for that round shall be adjusted (using 4.72 meters/ft. per second for rounds fired at 1146 Mils and 5.18 meters/ft. per second for rounds fired at 381 Mils to the mean velocity for the remaining rounds of that subgroup. If the outlier was high, then the correction is subtracted from the actual range.

$$r = \frac{V2 - V1}{Vi - (1 - Vi)}$$

Number of rounds	Outlier Criteria
10	.477
9	.512
8	.554

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4.4.2.2 Safety. The projectiles for this test shall be divided into two groups. Group 1 shall be conditioned and fired at 145°F - 5°F. For first three lot inspection, Group 1 shall consist of 10 projectiles and for normal inspection Group I shall consist of 6 projectiles. Group 2 shall be conditioned and fired at -65°F + 5°F. For first three lot inspection, Group 2 shall consist of 10 projectiles and for normal inspection Group 2 shall consist of 6 projectiles. The hot phase (Group 1) and the cold phase (Group 2) shall be fired at a quadrant elevation of 1146 Mils. Pressure data shall be recorded for information only.

4.4.2.3 Muzzle velocity and pressure and functioning. All projectiles shall be conditioned and fired at 70°F ± 2 1/2°F at Zone 8 using M200 series propelling charge conditioned at 70°F ± 2 1/2°F. The test shall be fired at a quadrant elevation of 1146 Mils.

5. PACKAGING.

5.1 Preservation packaging.

5.1.1 Level A.

5.1.1.1 Fiber container. The cartridges shall be preserved and packaged in accordance with dwg. 7549073 or 9258026.

5.2 Packing

5.2.1 Level A

5.2.1.1 Fiber container (dwg. 7549073) shall be packed in accordance with dwg. 7549072 or dwg. 9275181. Fiber container (dwg. 9258026) shall be packed in accordance with dwg. 9271850 or dwg. 9258027 as directed by Government Procuring Activity.

5.3 Marking

5.3.1 Marking of the wood box shall be in accordance with dwg. 7549072, 9258027, 9271850 or 9275180. The fiber container shall be marked in accordance with dwg. 7549073 or 9258026.

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6. NOTES

6.1 Intended use. The components covered by this specification are intended for use with 105MM Howitzer, M204.

6.2 Ordering data. See MIL-A-48078.

6.3 Drawings. Drawings listed in Section 2 of this specification under the heading U.S. Army Armament Research and Development Command (ARRADCOM) may also include drawings prepared by, and identified as, Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under the cognizance of ARRADCOM.

6.4 TNT Frothing. When TNT shows definite frothing tendencies, which may cause production of unsatisfactory charges, 0.1 percent maximum of sorbitan trioleate in accordance with Specification JAN-S-547 may be used. Projectile lots loaded with TNT which has been so treated should have data cards annoted accordingly.

6.5 Clean explosive scrap. Clean explosive scrap is defined as scrap formed in kettles, pails, risers and loading machines and used in the pouring and cooling operations, and scrap formed during the drilling operation.

6.6 Cost of check test. The contracting officer shall arrange for the contractor to be reimbursed for the expense incurred in the performance of the check tests. The test shall be conducted at Government expense without cost to the contractor who loaded the primers or to the contractor assembling the cartridge, and shall not constitute a basis for rejection against either contractor, except where deterioration has occurred as a direct result of carelessness in handling, storage, etc., permitted while the primer lot was under the jurisdiction of either contractor.

6.7 Inspection Equipment Designs. Inspection equipment designs are two types - Government Special Inspection Equipment (SIE) designs and contractor designs. SIE designs, if available, are designated by drawing numbers (on the Equipment Lists, if available, referenced on the Equipment Tabulation) or under the "Method of Inspection" headings in Section 4. Design responsiblity for all other inspection equipment is assigned to the contractor. Equipment to be designed by the contractor is designated "CD". The contractor need not furnish any design when a complete Government SIE design is provided. Unless otherwise specified, however, the contractor may submit alternate or modified contractor designs of SIE in accordance with 6.7.2 and 6.7.3 should he elect to do so.

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6.7.1 SIE designs. SIE designs may consist of any of the following:

a. Detailed drawings which completely depict all information necessary for the fabrication and use of the item of inspection equipment.
b. A source control drawing or a specification control drawing as defined in MIL-STD-100.
c. An envelope drawing, as defined in MIL-STD-100, which established the criteria which a detail design shall meet. When envelope drawings are specified, the contractor shall prepare designs which comply with the criteria therein.

6.7.2 Contractor designs. Contractor designs are required for all inspection equipment for which SIE designs are not specified and may include commercial equipment which the contractor proposes to use. (Commercial equipment is defined as unmodified equipment which is cataloged and available for purchase by the general public). Contractor designs shall include appropriate operating instructions, calibration procedures and maintenance procedures. Commercial equipment shall be fully described by catalog listings or other means which provide sufficient information to permit identification and evaluation by the Government and may include illustrations and engineering data. Designs shall be prepared for any special fixture(s) required to be used with commercial, or with SIE designs if not otherwise covered thereby (see 6.7.1c). Designs shall be as specified in the Contract Data Requirements Lists (DD Form 1423). This specification number, paragraph number and defect number from Section 4 shall be referenced on each contractor design together with the component or assembly drawing number and revision letter to which the specific design applies.

6.7.3 Submission of designs for approval. Contractor designs shall be approved by the Government prior to fabricating or procuring the equipment. Designs shall be submitted for approval in accordance with the stipulations, time frame and distribution specified in the Contract Data Requirements Lists (DD Form 1423) or in the contract. Partial submission of inspection equipment designs is permissible and encouraged. However, the completion date for design review will be based on the date of the final submission of designs and the required delivery schedule as stipulated in the contract. Submit designs as required, to: Commander, US Army Armament Research and Development Command, ATTN: DRDAR-QAR-I,

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Dover, New Jersey 07801. When the contractor submits inspection equipment designs to the Government for approval, he shall give the following information in his letter of transmittal:

a. The contract number.
b. The contract item (name, model number, etc.).
c. The designs remaining to be submitted and the expected date of submittal.

6.8 Radiographic procedures. Radiographic procedures and sample radiographs shall be submitted to the Commander, U.S. Army Armament Research and Development Command, Dover, New Jersey 07801, ATTN: DRDAR-QAR-Q for review and approval upon implementation of this Military Specification and subsequently at each 6 month period of production.

6.9 Photographic densities. "American Standard for Diffuse Transmission Density" published by American Standards Association, 70 East 40th Street, New York has been found satisfactory as a source of information for filter densities.

6.10 Submission of test data. In addition to the normal distribution of records, when the cartridge is procured by the Department of the Army, one copy of all ballistic test data will be forwarded to the following:

Commander
U.S. Army Armament Research and Development Command
ATTN: DRDAR-QAR-Q and DRDAR-LCA-G
Dover, New Jersey 07801

6.11 Standardization agreement. Certain provisions of this specification are the subject of international standardization agreement ABCA-Army-STD-56B. When amendment, revision, or cancellation of this specification is proposed, the department custodians will inform their respective Departmental Standardization Offices so that appropriate action may be taken respecting the international agreement concerned.

6.12 Definition of special defect. A special defect is a defect that judgement and experience indicate could, depending on the degree of variance from the design requirement, result in hazardous or unsafe conditions for individuals using, maintaining or depending upon the product.

6.13 Site of Inspection Lots. It is noted that the size of inspection lots of components assemblies or items of delivery may differ from the actual quantities contractually

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scheduled for delivery. However, in order to facilitate scheduling of tests or deliveries, inspection lots of items of delivery may be equivalent to such contract quantities provided all of the lot formation criteria and sampling provisions of this specification are maintained.

6.14 Informational control rounds. Informational control rounds are completed cartridges coming from one accepted lot of M1 projectiles and one accepted lot of M572 fuzes and one accepted lot of M200 propelling charges.

6.15 Submission of calibration and process control plans. Prior to implementation of calibration or process control plans in accordance with 4.3.2, the plans shall be approved by U.S. Army Armament Research and Development Command, ATTN: DRDAR-QAR-Q, Dover, New Jersey 07801.

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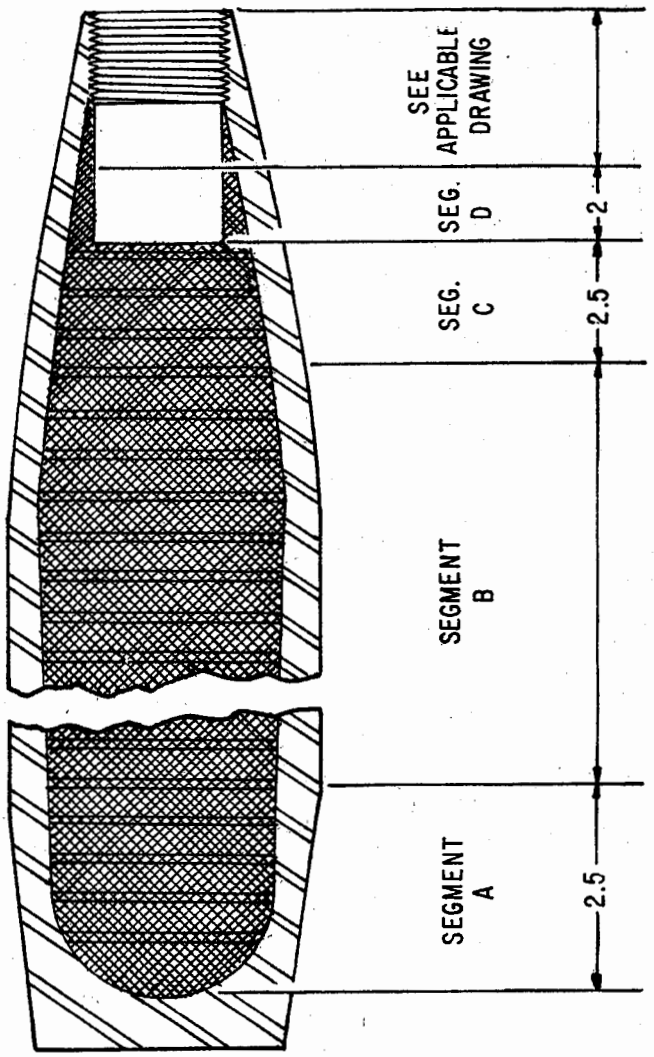
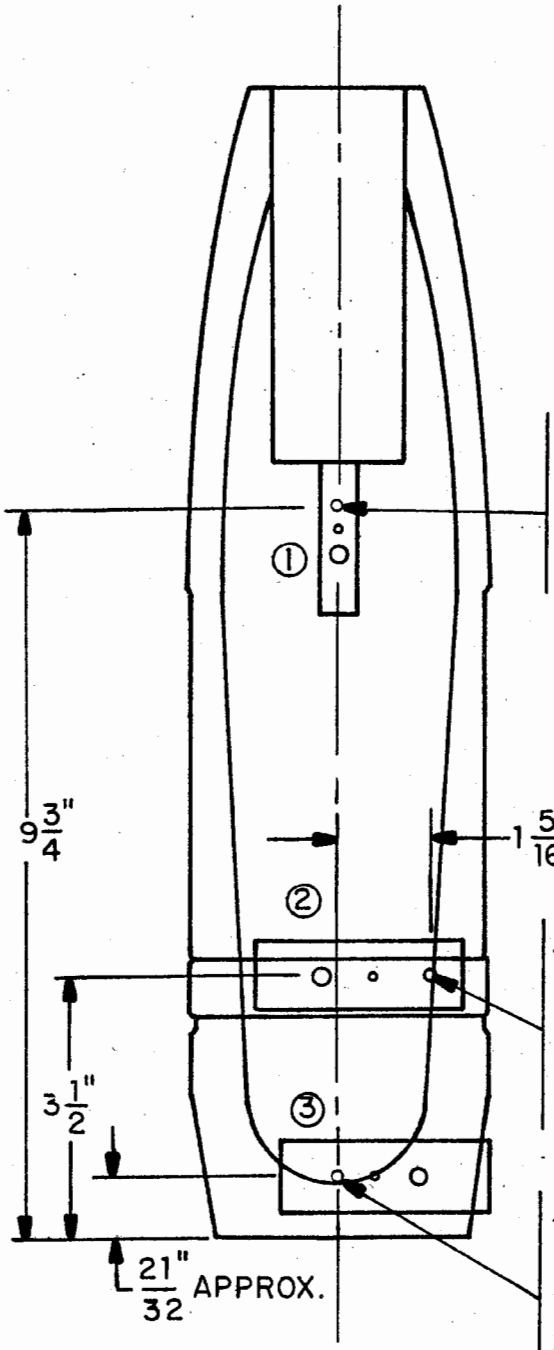


FIGURE 1

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PROJECTILE 105MM, HE, TNT, M1
RADIOGRAPHIC PENETRAMETER REQUIREMENTS



Penetrameters Constructed Per MIL-STD-453
Dimensions shown are for location of center 2T holes.

PENETRAMETER FOR 1.5" STEEL
.75 Total steel wall plus 3.375" explosive.
1.5" equivalent thickness
2% Penetrameter thickness =.030"
2T Hole diameter =.060"

PENETRAMETER FOR 2.7" STEEL
2.75" Total Steel wall plus .82" explosive at 1/16" from tangent point.
2.74" equivalent thickness
2% penetrameter thickness =.054"
2T Hole diameter =.108"

PENETRAMETER FOR 3.1" STEEL
3.0" Total steel wall plus .75" explosive at 1/16" from tangent point.
3.15" equivalent thickness
2% penetrameter thickness =.062"
2T Hole diameter =.124

FIGURE 2

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6.16 Proving ground test summary (zone firings).

Test	Temp. (F)	Quadrant Elevation (Mils)	Sample size/group (tightened inspec.)	Sample size/group (Normal inspec.)
1. Safety (w/M572 Fuze)	+145	1146	10	6
	- 65	1146	10	6
* 2. Functioning/ Precision (w/M572 Fuze)	70	1146	NA	Two groups of ten
	70	381	NA	One group of ten
** 3. Informational Control rds. (for Funct/Precision firings)	70	1146	NA	4
	70	381	NA	4
4. Functioning (w/M572 Fuze)	70	1146	20	10
5. Muzzle Velocity & Pressure (w/M572 Fuze)	70	1146	Same cartridges as 4. above.	

* This test shall be performed until three consecutive lots are accepted.

** These firings are only needed until three consecutive lots are accepted.

6.17 Factors. The factors, as taken from Table A-21 of AMC Pamphlet 706-114, using the 95 percent upper confidence limit (A.05) and n-1 degrees of freedom, make allowance for the probability that a standard deviation from a sample exceeds the true standard deviation by chance alone. The factor for a normal 20 round group is 0.79, and the factor for a normal 10 round group is 0.73.

Custodian:
Army-AR

Preparing Activity:
Army-AR

Project No.: 1315-A164

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